

Beria Ayşenur Can

Particle Physicist | Data Scientist & Engineer

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beriaacan

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personal website

Profile

- Graduate researcher at ITU and Member of the CMS Collaboration at CERN, specializing in the HGCal project. Expert in bridging high-energy physics with industrial-grade data engineering. Proficient in developing high-throughput analysis pipelines, applying deep learning to massive LHC datasets, and optimizing detector physics simulations. Leveraging a dual background in Data Science and Physics to implement scalable, production-ready computational solutions for fundamental research.

Education

• Istanbul Technical University

M.Sc. in Physics Engineering

2024 – Present

Istanbul, Türkiye

Specialization in particle physics & computational physics

• Istanbul Technical University

B.Sc. in Physics Engineering

2020 – 2024

Istanbul, Türkiye

Graduated **3rd in department**; GPA: 3.3/4.0

Thesis: **Investigation of Radon Particle Formation and Measurement for Earthquake Prediction Using Deep Learning**

Focus on computational physics, machine learning, simulation techniques.

Experience

• CERN (CMS Experiment)

Cern User / Project Associate (HGCal)

2025 – Present

Geneva, Switzerland

- Contributing to the **High Granularity Calorimeter (HGCal)** upgrade, focusing on the integration and **quality assurance** of **optical fiber data transmission** systems.
- Facilitating **hardware-to-software data flow** consistency, ensuring high-speed **signal integrity** for high-luminosity LHC operations.
- Collaborating within the **CMS analysis team** to align detector instrumentation with computational physics requirements.

• Adin.ai

Data Engineer & Scientist

2025 – Present

Istanbul, Türkiye

- Developing and deploying scalable **REST APIs** using **FastAPI** to provide real-time marketing analytics.
- Implementing complex **statistical research** and metric calculations (**CPM, CPR, Nielsen**) using **NumPy** and **Pandas** to evaluate advertising performance.
- Building and maintaining robust **data pipelines** using **SQL (PostgreSQL/DBEaver)** and **Python**, ensuring seamless data flow and **visualization**.
- Utilizing **Postman** for API testing and integrating **cloud-based workflows** within **AWS** environments (**Cognito, S3**) for high-concurrency requests.

• SAS

Associate Analytical Consultant / Technical Intern

2024

Istanbul, Türkiye

- Developed **Machine Learning** models for **Fraud Detection** using **Python, SQL, and Scikit-learn**.
- Performed **anomaly detection** and predictive analytics to identify fraudulent patterns, enhancing financial security protocols.
- Conducted **model validation** and result interpretation, translating complex analytical findings into actionable business insights.

• TUBITAK 1001 Project

Research Student (Scholar)

2023 – 2024

Istanbul, Türkiye

- Analyzed radon concentration data; applied **time-series modeling** and **Deep Learning** to investigate seismic correlations.
- Managed the **end-to-end data pipeline**, including acquisition, **noise reduction**, and visualization of environmental sensor data.

Technical Skills

- **Programming:** Python, C++, SQL, Matlab, Bash
- **Machine Learning & Data Science:** Scikit-learn, NumPy, Pandas, SciPy, Anomaly Detection, Time-Series Forecasting
- **Data Engineering & Web:** FastAPI, REST APIs, ETL Pipelines, DBeaver, Postman
- **Physics & Analytics:** ROOT (CERN), COMSOL, SimFlow
- **Cloud & Tools:** AWS (Cognito/S3/EC2), Linux, Git, Docker, Matplotlib/Seaborn (Visualization)